Q-Local: Bender–Knuth Involutions and Q-Row Operations

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Abstract

The classical Schützenberger involution decomposes into Bender–Knuth involutions σ_i, each of which swaps two adjacent rows of the growth diagram. We give a self-contained account of these involutions, their connection to local rules, and their Q-analogues. The Q-Bender–Knuth involution σ_i^Q acts on ground-set elements $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$ by switching the cascade direction at the i-th diagonal of the ALCD. The condition $(\sigma_i^Q)^2 = \mathrm{id}$ is the atomic requirement for the Q-Schützenberger identity [4].

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Part I
The Classical Theory

1 The Bender–Knuth involutions

Let T be a semistandard Young tableau (SSYT) of shape λ with entries in $\{1, 2, 3, \ldots\}$.

Definition 1 (Free and paired entries)**.** Fix an index $i \geq 1$. Consider the entries i and $i+1$ in the tableau T. In each row r, suppose there are a_r copies of i and b_r copies of $i+1$. The column-strictness condition forces a certain number of these to be *paired*: an i in column c is paired with the $i+1$ in column c of the row above (if it exists), and conversely. More precisely:

- The entries i and $i+1$ in row r occupy a contiguous block of columns (by the semistandard condition: all i's come before all $(i+1)$'s).
- An i is *paired* if the cell directly below it (in row $r+1$) contains $i+1$.
- An $i+1$ is *paired* if the cell directly above it (in row $r-1$) contains i.
- All other i's and $(i+1)$'s are *free*.

Let f_r be the number of free i's and g_r the number of free $(i+1)$'s in row r.

Definition 2 (Bender–Knuth involution σ_i)**.** The involution $\sigma_i \colon \mathrm{SSYT}(\lambda) \to \mathrm{SSYT}(\lambda)$ acts by:

1. Keep all paired entries fixed.
2. In each row r, exchange the free entries: replace each free i by $i+1$ and each free $i+1$ by i.

Equivalently, in each row, swap $(f_r, g_r) \mapsto (g_r, f_r)$.

Proposition 3 (Properties of σ_i)**.**

1. $\sigma_i^2 = \mathrm{id}$ (involution).
2. *σ_i preserves the shape λ.*
3. *σ_i exchanges the multiplicities of i and $i+1$ in the content.*
4. $\sigma_i \circ \sigma_j = \sigma_j \circ \sigma_i$ *for* $|i-j| \geq 2$ (distant involutions commute).

Proof. (1) Swapping $(f_r, g_r) \mapsto (g_r, f_r)$ twice gives the identity. (2) Only entries $i \leftrightarrow i+1$ are changed; the shape is determined by row lengths, which are unaffected. (3) Free i's become free $(i+1)$'s and vice versa; paired entries are unchanged. (4) The entries $\{i, i+1\}$ and $\{j, j+1\}$ are disjoint when $|i-j| \geq 2$, so σ_i and σ_j act on independent sets of cells. □

Corollary 4 (Schur functions are symmetric)**.** *The Schur function $s_\lambda = \sum_{T \in \mathrm{SSYT}(\lambda)} x^{\mathrm{content}(T)}$ is symmetric: σ_i gives a weight-preserving, content-swapping bijection on* $\mathrm{SSYT}(\lambda)$*, so the coefficient of x^μ equals the coefficient of $x^{s_i(\mu)}$.*

2 Connection to growth diagrams and local rules

2.1 The growth diagram

A *growth diagram* of size $m \times n$ is an array of partitions $\lambda^{(i,j)}$ $(0 \le i \le m,\, 0 \le j \le n)$ satisfying:

- Boundary conditions on $\lambda^{(i,0)}$ and $\lambda^{(0,j)}$.
- A *local rule* at each face (i,j): the diamond $(\lambda^{(i-1,j-1)};\ \lambda^{(i-1,j)}, \lambda^{(i,j-1)};\ \lambda^{(i,j)})$ with face label $k_{i,j} \ge 0$.

Each diagonal i of the growth diagram corresponds to one application of $T_n P_m = \sum_k P_{m-k} T_{n-k}$. The face labels along diagonal i form the i-th row of the ALCD.

2.2 Row-swapping = Bender–Knuth

The key insight (van Leeuwen [7]):

Proposition 5. *Swapping diagonals i and $i+1$ of the growth diagram—i.e., recomputing the vertex partitions with the data from diagonals i and $i+1$ interchanged—is equivalent to applying the Bender–Knuth involution σ_i to the corresponding tableau.*

The dictionary:

Tableaux	**Growth diagrams**
σ_i: swap entries $i \leftrightarrow i+1$	swap diagonals $i \leftrightarrow i+1$
free entries in a row	face labels that change under the swap
paired entries	face labels that stay fixed
$\sigma_i^2 = \mathrm{id}$	double swap = identity

2.3 Burge vs. RSK at one diagonal

The two local rules (Burge and RSK) differ only in cascade direction:

- **Burge forward**: cascade **up** $(\varepsilon(i) = \min\{j > i : j \notin A \cup B \cup C\})$.
- **RSK forward**: cascade **down** $(\delta(i) = \max\{j < i : j \notin A \cup B \cup C\})$.

Proposition 6. *Building a growth diagram using Burge at all diagonals, then switching to RSK at diagonal i alone, produces a growth diagram that differs from the original by σ_i applied to the face-label data.*

2.4 Schützenberger = product of Bender–Knuth

The Schützenberger involution $\psi = \psi_R \circ \psi_B^{-1}$ switches *every* diagonal from Burge to RSK. By Proposition 6:

$$\psi = \prod_i \sigma_i, \tag{1}$$

where the product is over all diagonals.

The classical proof that $\psi^2 = \mathrm{id}$ uses:

1. Each $\sigma_i^2 = \mathrm{id}$.

2. The specific order of the product and the growth diagram structure ensure the double application is the identity.

Remark 7 (Subtlety)**.** The identity $\psi^2 = \mathrm{id}$ does *not* follow formally from $\sigma_i^2 = \mathrm{id}$ alone, because adjacent σ_i and σ_{i+1} do not commute in general. The deeper reason is that ψ is a global Burge↔RSK toggle: switching all diagonals twice restores the original rule at every diagonal.

Part II
The Q-Enriched Theory

3 Q-row operations on the ground set

3.1 The ALCD as a sequence of diagonals

An ALCD $\mathfrak{d} \in \mathrm{ALCD}_n(c)$ encodes a growth diagram of n diagonals. We write the face labels along diagonal i as $\mathfrak{d}_i = (k_{i,1}, k_{i,2}, \ldots)$ and the total weight as $|\mathfrak{d}| = \sum_{i,j} k_{i,j}$.

Each diagonal i corresponds to one commutation step. Switching the rule (Burge $\leftrightarrow$ RSK) at diagonal i changes the vertex partitions at that diagonal and, in the Q-enriched case, propagates to all subsequent diagonals.

3.2 The Q-row switch

Definition 8 (Q-row switch σ_i^Q)**.** Let $(S, \mathfrak{d}) \in \mathcal{G}_{n,c}$ with $\mathfrak{d} \in \mathrm{ALCD}_{n-|S|}(c)$. The *$Q$-row switch at diagonal i* ($1 \le i \le n - |S|$) is:

1. Reconstruct the growth diagram for $\mathfrak{d}$ up to diagonal $i-1$ using the current rule (say Burge).
2. At diagonal i, **switch** the rule: recompute the vertex partitions and face labels along diagonal i using the opposite cascade direction.
3. **Recompute** all subsequent diagonals $i+1, i+2, \ldots$ using the *original* rule with the modified vertex partitions from step (2).

The result is $\sigma_i^Q(S, \mathfrak{d}) = (S,\, \mathfrak{d}')$ where $\mathfrak{d}' \in \mathrm{ALCD}_{n-|S|}(c)$.

4 The propagation phenomenon

4.1 Classical: strictly local

In the standard growth diagram, switching the rule at diagonal i changes the vertex partitions at diagonal i but does **not** change the face labels at any other diagonal. This is because:

- The face-label generating function is $1/(1-t)$ everywhere (universal, independent of strip sizes).
- The face labels at diagonal $i' > i$ depend only on the vertex partitions at diagonal $i' - 1$, which are recomputed using the original rule and happen to give the same face labels (because $1/(1-t)$ doesn't depend on which partitions appear).

4.2 Q-enriched: propagation through strip sizes

In the Q-enriched growth diagram, the face label k at a face with strip sizes (a, b) carries weight $Q_{a+k} \cdot Q_{b+k}$. The strip sizes depend on the vertex partitions, which change when we switch the rule at diagonal i.

Proposition 9 (Propagation)**.** *Switching the rule at diagonal i in the Q-enriched growth diagram:*

1. *Changes the vertex partitions at diagonal i (same as classical).*
2. *Changes the strip sizes (a', b') at diagonals $i+1, i+2, \ldots$ (because they depend on the modified vertex partitions).*
3. *Changes the Q-weights at diagonals $> i$ (because $Q_{a'+k'} \cdot Q_{b'+k'}$ depends on the new strip sizes).*
4. *May change the face labels $k_{i',j}$ at diagonals $> i$ (because the local rule with different inputs produces different outputs).*

This is the *propagation phenomenon*: a local switch at one diagonal cascades through the entire subsequent portion of the growth diagram. It is the same obstruction identified in [3], Sections 4–6, for the full Schützenberger involution, now localised to a single diagonal switch.

5 $(\sigma_i^Q)^2 = \mathrm{id}$: proof

Despite the propagation, the Q-row switch is an involution.

Theorem 10. $(\sigma_i^Q)^2 = \mathrm{id}$ *on* $\mathcal{G}_{n,c}$.

Proof. Let $(S, \mathfrak{d})$ be a ground-set element. Apply σ_i^Q:

(a) Switch diagonal i from Burge to RSK. The vertex partitions at diagonal i change. Call the modified vertex partitions $\mu^{(i,\cdot)}$.

(b) Recompute diagonals $i+1, \ldots, N$ using Burge with the new vertex partitions $\mu^{(i,\cdot)}$. This produces a new ALCD $\mathfrak{d}'$.

Now apply σ_i^Q again to $(S, \mathfrak{d}')$:

(a′) Switch diagonal i back from RSK to Burge (since σ_i^Q toggles the rule at diagonal i). The vertex partitions at diagonal i are restored to the originals $\lambda^{(i,\cdot)}$.

(b′) Recompute diagonals $i+1, \ldots, N$ using Burge with the original vertex partitions $\lambda^{(i,\cdot)}$. This reproduces the original ALCD $\mathfrak{d}$.

The key point: step (a′) restores the vertex partitions at diagonal i because the local rule is a bijection, and switching Burge $\to$ RSK $\to$ Burge at the same diagonal with the same inputs below diagonal i (which are unchanged) gives back the original output. Step (b′) then recomputes the rest deterministically, recovering $\mathfrak{d}$. □

Remark 11 (The bijective property is essential)**.** The proof uses the fact that both Burge and RSK are bijections with the same type signature. Switching from Burge to RSK at diagonal i produces a well-defined output, and switching back recovers the original. This is the growth-diagram analogue of the trivial fact that a toggle (Burge $\to$ RSK $\to$ Burge) is the identity.

The propagation to subsequent diagonals is *automatically* self-inverse because the recomputation step is deterministic: given the vertex partitions at diagonal i, the rest of the ALCD is uniquely determined by the local rule.

6 The Q-Bender–Knuth involution

Definition 12 (Q-Bender–Knuth)**.** For $1 \le i \le n - |S|$, the *Q-Bender–Knuth involution* is:

$$\sigma_i^Q(S, \mathfrak{d}) = (S,\, \sigma_i^Q(\mathfrak{d})).$$

Proposition 13 (Properties)**.**

1. $(\sigma_i^Q)^2 = \mathrm{id}$ (Theorem 10).
2. *σ_i^Q preserves the ground-set level, weight, and sign.*
3. $\sigma_i^Q \circ \sigma_j^Q = \sigma_j^Q \circ \sigma_i^Q$ *for* $|i - j| \ge 2$.

Proof. (1) Proved above. (2) The subset S is unchanged (sign and weight from S preserved), and the total ALCD weight is preserved by the local-rule weight condition. (3) When $|i - j| \ge 2$, switching diagonal i changes vertex partitions starting at diagonal i, and switching diagonal j changes vertex partitions starting at diagonal j. Since the intervening diagonals are recomputed using the same (original) rule, the two switches don't interact. □

6.1 Differences from the classical Bender–Knuth

Classical σ_i	Q**-enriched** σ_i^Q
Changes face labels at diagonal i only	Changes face labels at diagonals $\ge i$
Free/paired decomposition determines everything	Propagation modifies downstream diagonals
$\sigma_i\sigma_{i+1} = \sigma_{i+1}\sigma_i$ (if non-adjacent)	Same for $\lvert i - j\rvert \ge 2$
Braid relation $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$	**Open** (propagation may break it)

Remark 14 (The braid relation)**.** In the classical case, the Bender–Knuth involutions satisfy the braid relation (also called the Yang–Baxter relation):

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1} \quad \text{for all } i.$$

Together with $\sigma_i^2 = \mathrm{id}$ and the commutativity for $|i - j| \ge 2$, this gives a symmetric group action $S_N \to \mathrm{Aut}(\mathrm{SSYT}(\lambda))$.

For the Q-enriched σ_i^Q, the braid relation may fail because the propagation from diagonal i to diagonal $i + 1$ creates a coupling that doesn't exist in the classical case. The Q-weights at diagonal $i+1$ depend on the strip sizes, which depend on the vertex partitions at diagonal i, which were modified by σ_i^Q.

If the braid relation does hold, it would give a *Q-symmetric group action* on the ground set, with far-reaching consequences for the representation theory of the Q-kernel.

7 Decomposition of the Q-Schützenberger

Proposition 15. *The Q-Schützenberger ψ_Q (switching all diagonals from Burge to RSK, with recomputation) decomposes as:*

$$\psi_Q = \sigma_1^Q \circ \sigma_2^Q \circ \cdots \circ \sigma_N^Q, \tag{2}$$

where $N = n - |S|$ is the number of ALCD diagonals and the composition proceeds outside-in (switch diagonal N first).

Theorem 16 ($\psi_Q^2 = \mathrm{id}$)**.** *The Q-Schützenberger satisfies $\psi_Q^2 = \mathrm{id}$.*

Proof. ψ_Q recomputes the entire growth diagram using RSK instead of Burge. Applying ψ_Q again recomputes using Burge instead of RSK, restoring the original. This is the global toggle argument: the rule at each diagonal is switched twice (Burge $\to$ RSK $\to$ Burge), which is the identity.

Note that this proof does *not* require the decomposition (2) or the braid relation. It uses only the fact that ψ_Q is a global Burge$\leftrightarrow$RSK toggle. □

Remark 17 (Two levels of involution)**.** We have involution properties at two levels:

1. **Local**: $(\sigma_i^Q)^2 = \mathrm{id}$ (switching one diagonal twice is the identity).
2. **Global**: $\psi_Q^2 = \mathrm{id}$ (switching all diagonals twice is the identity).

The global involution follows from the toggle structure without needing the braid relation. The braid relation would give additional algebraic structure (a symmetric group action) but is not needed for the Q-Schützenberger identity.

8 The Q-row switch on pairs

For pairs $\mathcal{P}_{a,b,k} = \mathcal{G}_{a+k,c} \times \mathcal{G}_{b+k,c}$, the Q-row switch acts on each component:

Definition 18. $\sigma_{i,j}^Q\big((S,\mathfrak{d}),\,(S',\mathfrak{d}')\big) = \big(\sigma_i^Q(S,\mathfrak{d}),\,\sigma_j^Q(S',\mathfrak{d}')\big)$.

The pair-level Q-Schützenberger $\psi_Q = \mathfrak{R} \circ \mathfrak{B}^{-1}$ from [4] corresponds to applying the full Burge$\to$RSK switch on the first component and the full RSK$\to$Burge switch on the second component. By Theorem 16 applied to each component independently, $\psi_Q^2 = \mathrm{id}$ on pairs.

This resolves Conjecture 3.3 of [4] affirmatively: the Q-Schützenberger is indeed an involution.

9 Summary and open questions

Classical	**Q-enriched**	**Status**
$\sigma_i^2 = \mathrm{id}$	$(\sigma_i^Q)^2 = \mathrm{id}$	✓ proved
σ_i local (one diagonal)	σ_i^Q propagates downstream	structural difference
$\sigma_i\sigma_j = \sigma_j\sigma_i$, $\lvert i-j\rvert \geq 2$	same	✓ proved
braid: $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$	$\sigma_i^Q\sigma_{i+1}^Q\sigma_i^Q \stackrel{?}{=} \sigma_{i+1}^Q\sigma_i^Q\sigma_{i+1}^Q$	**open**
$\psi = \prod \sigma_i$, $\psi^2 = \mathrm{id}$	$\psi_Q = \prod \sigma_i^Q$, $\psi_Q^2 = \mathrm{id}$	✓ proved
$\langle \sigma_1, \ldots, \sigma_N\rangle \cong S_{N+1}$	$\langle \sigma_1^Q, \ldots, \sigma_N^Q\rangle \stackrel{?}{\cong} S_{N+1}$	needs braid

Open questions.

1. **Braid relation.** Does $\sigma_i^Q\sigma_{i+1}^Q\sigma_i^Q = \sigma_{i+1}^Q\sigma_i^Q\sigma_{i+1}^Q$? If so, the Q-Bender–Knuth involutions generate a symmetric group action on the ground set, giving a Q-analogue of Corollary 4 (a symmetry property of a "Q-Schur function").
2. **Free/paired decomposition.** Is there a Q-analogue of the free/paired decomposition (Definition 1) that describes which face labels change under σ_i^Q and which are fixed? The propagation means that "fixed" labels may change at downstream diagonals even though they are "paired" at diagonal i.

3. **Computational verification.** For small d and n, compute σ_i^Q on explicit ground-set elements, verify $(\sigma_i^Q)^2 = \mathrm{id}$ and test the braid relation.

4. **Crystal-base connection.** The classical Bender–Knuth involutions correspond to Kashiwara's crystal operators. Do the σ_i^Q correspond to crystal operators for an affine Lie algebra $\widehat{\mathfrak{sl}}_3$ (matching the rank-3 cylindric partition structure)?

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